1. Describe one result you found interesting doing your task.

One thing I found interesting was that our group ended up having a shared interest in both HPC and machine learning rather than needing to immediately split into separate groups. After discussing possible directions and looking at related research, we became interested in using the HPC resources available to us to train or fine-tune an ML model for modeling turbulence in tokamak fusion reactors. The papers we found were especially interesting because they showed that machine learning has already been used to model turbulent plasma behavior, which gave us a better idea of how our project could connect HPC, ML, and fusion research.

2. Describe one thing you struggled with doing your task.

The biggest difficulty was deciding on a specific project direction without having enough context yet about the exact problem, data, and expected outcome. Because we are still learning about fusion and plasma turbulence, it was difficult to make concrete decisions about things like our research question, what type of model we would use, or what journal we might eventually target.

3. Identify one choice that you made in your investigation. Why did you make this choice?

One choice we made was to not divide into separate project groups yet and instead continue exploring an HPC/ML-focused direction together. We felt that committing to a very specific topic or publication goal this early would be premature. Instead, we chose to find and share relevant papers so that we can build more background knowledge before narrowing our project scope. Another reasonable choice would have been to split into groups immediately based on publication goals or different technical areas, but we felt we needed more information first.